# Compact Single-Turn Coupled Inductor for MHz-Range Buck Converters with Zero-Voltage Switching

Adrian Keil[1], Gean Maia de Sousa[2], Vincent Lorentz[3], Thomas Heckel[1], Marcelo Lobo Heldwein[2]
1 Fraunhofer Institute for Integrated Systems and Device Technology IISB, Erlangen, Germany
2 Chair of High-Power Converter Systems, Technical University of Munich, Germany
3 Chair of Electronics for Electrical Energy Storage, University of Bayreuth, Germany

***Abstract*—The next generation of automotive vehicles and data centers requires highly compact and efficient 48 V to 12 V point-of-load converters. This paper presents a novel single-turn planar inductor geometry with four poles and dual air gap for operation beyond 1 MHz that minimizes copper losses from external proximity effect. An experimental 1 kW prototype achieves a power density of 80 kW/L (1300 W/in³) and a peak efficiency of 96.3%, demonstrating the potential of the inductor structure.**

***Index Terms*—automotive 48 V to 12 V converters, coupled inductor, high power density, soft switching**

## I. Introduction

With a growing power demand, power distribution in both conventional and electric vehicles poses an increasing challenge. Traditionally, 12 V are used to distribute the power to all auxiliary devices which requires large cable diameters. Moving to a 48 V distribution bus reduces wiring cost and conduction losses [1]. Since low-power devices still operate at 12 V, highly compact and efficient point-of-load converters are required. This conversion stage is a critical part of distributed power architectures, and its performance has a direct impact on system-level efficiency, thermal design, and overall build volume [1]. With the rise of Gallium Nitride (GaN) power devices, operating converters in the MHz-range has become feasible, significantly reducing their size [2]. At these frequencies, using planar magnetics with printed circuit board (PCB) windings is beneficial as they allow elaborate winding structures, direct integration with the other circuitry, and efficient cooling due to their large surface area [3]–[6]. Resonant converters with fixed gain based on planar transformers are a common choice for 48 V to 12 V conversion, but they are unsuitable when regulation over a wide input voltage range is required. In these cases, the classical buck converter is appropriate but the design of compact and efficient planar inductors is challenging due to the due to the inability to use interleaved windings, the fringing effect of the air gap, and the DC-bias in the core [7]–[10].

In this paper, a 1 kW, 2-phase coupled inductor buck converter for 48 V to 12 V conversion is studied. To minimize the overall volume, a target switching frequency range of 1 MHz to 3 MHz was selected. Compared to previous work [3], [11]–[14], this converter's low operating voltage requires a very low inductance. The design becomes particularly challenging due to the large phase current of 40 A with 100 A of ripple which is required to achieve zero-voltage switching. Firstly, the effects of coupling on the electrical parameters are analyzed mathematically. Afterwards, different geometries for the single-turn inductor are optimized using a novel winding geometry and dual air gaps. The most promising geometry based on the four-pole structure is implemented and tested in an experimental prototype.

## II. Coupled inductor buck converter in TCM

*a) Working principle:* The topology of the two phase coupled inductor buck converter is shown in Fig. 1. It operates like a conventional two-phase buck with both legs switched 180° out of phase. When leg 2 is pulled high, the current in leg 1 is also affected as shown in Fig. 2. For positive coupling, the falling slope is steepened while for negative coupling it is flattened. Operation in triangular current mode (TCM) enables zero-voltage switching (ZVS) by increasing the phase current ripple to more than twice the average phase current (Fig. 2) [15]. The low-side switch is kept on after the zero-crossing of the leg current for a short period of time such that the current becomes negative. Once the low-side switch is turned off, the inductor current charges/discharges the output capacitances until the high-side GaN-FET enters reverse conduction. High losses are observed during reverse conduction, which requires precise dead-time calculation and adaptive switching frequency [16]. Turn-off losses remain, but are small due to the high output capacitance.

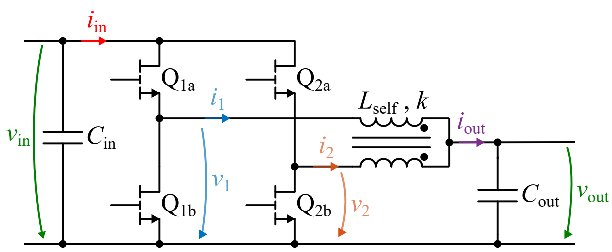
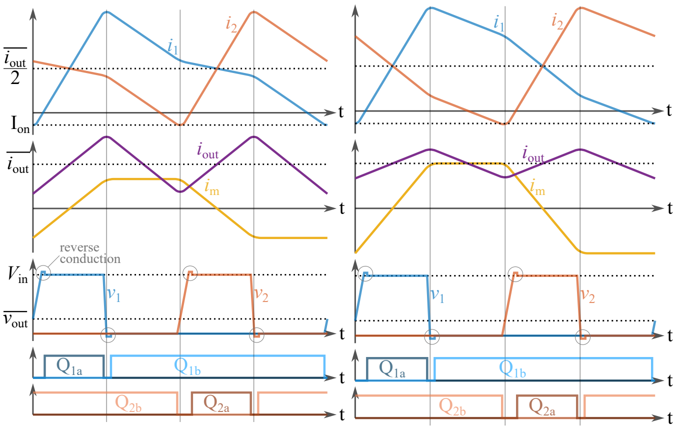


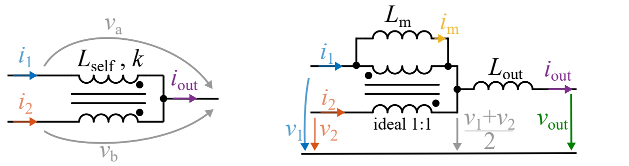

Fig. 1: Schematic of the Coupled Inductor Buck Converter.

(C)2026IEEJ




(a) Negative Coupling (b) Positive Coupling

Fig. 2: Waveforms for positive and negative coupling; dead-times are exaggerated. It can be seen that the current slopes change but the behavior in the vicinity of the switching instances is fundamentally the same.




(a) Coupled Inductor (b) Transformer Model

Fig. 3: The coupled inductor can be described with a self inductance $L_{\text{self}}$ and coupling factor $k$ or by using an equivalent circuit consisting of an ideal 1:1 transformer with magnetizing inductance $L_{\text{m}}$ and common output inductance $L_{\text{out}}$.

*b) Impact of the Coupling Factor:* The symmetrical coupled inductor shown in Fig. 3 (a) consists of two identical coils. They are wound in a way, that the flux of one coil links with the flux of the second coil and vice versa with both coils connected on one side. This can be described mathematically using

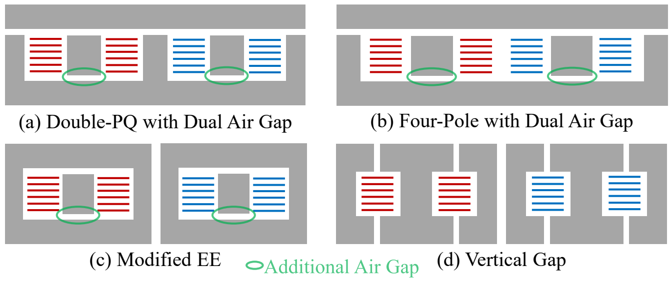
$$\begin{bmatrix} v_{\text{a}} \\ v_{\text{b}} \end{bmatrix} = \begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} L_{\text{self}} \begin{bmatrix} \frac{di_1}{dt} \\ \frac{di_2}{dt} \end{bmatrix} \tag{1}$$

with self inductance $L_{\text{self}}$ and magnetic air gap $k$ ($-1 \leq k \leq 1$). In order to provide a more intuitive understanding, the equivalent circuit in Fig. 3 (b) is introduced. This model is not very common in the analysis of coupled inductor circuits, but has one major advantage: The currents of the two inductors only depend on the input and output voltages. The equivalency of both models can be shown from the governing equations:

$$\begin{bmatrix} v_{\text{a}} \\ v_{\text{b}} \end{bmatrix} = \begin{bmatrix} L_{\text{m}} + L_{\text{out}} & -L_{\text{m}} + L_{\text{out}} \\ L_{\text{m}} + L_{\text{out}} & L_{\text{m}} + L_{\text{out}} \end{bmatrix} \begin{bmatrix} \frac{di_1}{dt} \\ \frac{di_2}{dt} \end{bmatrix} \tag{2}$$

Comparing the coefficients shows that both models are identical for

$$\begin{aligned} L_{\text{out}} &= (1+k)\,\frac{L_{\text{self}}}{2} \\ L_{\text{m}} &= (1-k)\,\frac{L_{\text{self}}}{2}. \end{aligned} \tag{3}$$




(a) Double-PQ with Dual Air Gap (b) Four-Pole with Dual Air Gap (c) Modified EE (d) Vertical Gap

Fig. 4: Structure of the studied cores in side view if cut through the middle. First winding in red, second winding in blue.

The voltage at the virtual central node in the transformer model is only dependent on the two leg voltages $v_1$ and $v_2$, decoupling the governing equations:

$$\begin{aligned} \frac{\mathrm{d}i_{\text{out}}}{\mathrm{d}t} &= \frac{1}{L_{\text{out}}}\left(\frac{v_1+v_2}{2} - v_{\text{out}}\right) \\ \frac{\mathrm{d}i_{\text{m}}}{\mathrm{d}t} &= \frac{1}{L_{\text{out}}}\left(\frac{v_1-v_2}{2}\right) \\ \text{with} \quad i_{\text{out}} &= i_1 + i_2 \quad \text{and} \quad i_{\text{m}} = i_1 - i_2 \end{aligned} \tag{4}$$

From this, the differential equations for each interval can be easily calculated and equations for the important converter parameters can be derived. An effective duty cycle $D_{\text{eff}}$ is introduced with $D_{\text{eff}} = D$ for $D \leq 0.5$ and $D_{\text{eff}} = 1 - D$ for $D > 0.5$ to create universal equations. The peak-to-peak output ripple is then given by

$$\Delta I_{\text{out}} = \frac{2\,V_{\text{in}}}{f_{\text{s}}\,(1+k)\,L_{\text{self}}} D_{\text{eff}}\left(\frac{1}{2} - D_{\text{eff}}\right). \tag{5}$$

The peak-to-peak ripple in each leg which is important for soft switching is

$$\Delta I_{\text{leg}} = \frac{V_{\text{in}}\,D_{\text{eff}}}{2\,f_{\text{s}}\,L_{\text{self}}}\left(\frac{2}{1+k}\left(\frac{1}{2} - D_{\text{eff}}\right) + \frac{1}{1-k}\right). \tag{6}$$

$I_{\text{on}}$ needs to be below a certain negative value $I_{\text{on,max}}$ to guarantee a sufficiently short dead-time. This is fulfilled for

$$f_{\text{s}} < \frac{V_{\text{in}}\,D_{\text{eff}}}{2\,L_{\text{self}}\,(\overline{i_{\text{out}}} + 2\,I_{\text{on,max}})}\left(\frac{2}{1+k}\left(\frac{1}{2} - D_{\text{eff}}\right) + \frac{1}{1-k}\right) \tag{7}$$

where $\overline{i_{\text{out}}}$ is the average value of $i_{\text{out}}$.

## III. Inductor Design

*a) Inductor geometry:* This work introduces an additional air gap to different designs, as shown in Fig. 4. Four geometries, two coupled, and two uncoupled ones, were compared: The Double-PQ structure (Fig. 4 (a)) was originally introduced by [3] and combines two EQ-cores placed next to each other. The four-pole structure (Fig. 4 (b)) is very similar but omits the central pole [14]. Both of these geometries were originally constructed with two core pieces, such that there is only a single air gap at the top. The third geometry (Fig. 4 (c)) is basically an EE core but instead of a single central gap, two gaps are used. Lastly, the vertical gap geometry (Fig. 4 (d)) proposed in [5] is investigated. The vertical air gap creates

(C)2026IEEJ

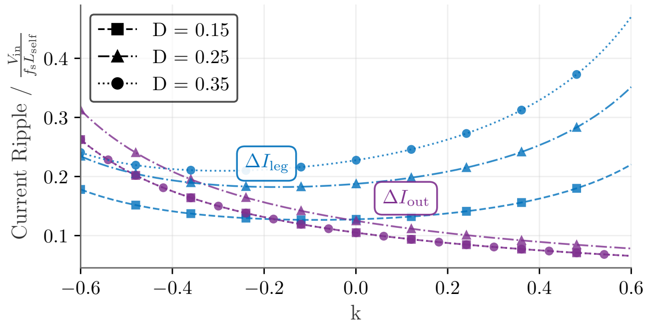
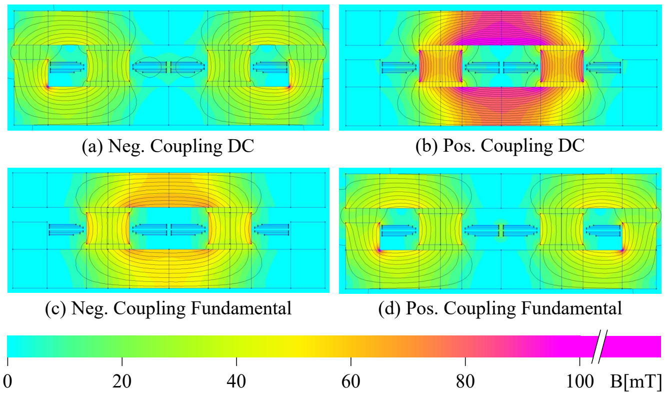



Fig. 5: Peak-to-peak leg and output current ripple in dependency of the coupling factor for different duty cycles $D$, normalized to $V_{\mathrm{in}}/(f_{\mathrm{s}}\,L_{\mathrm{self}})$. *Note:* For $D = 0.15$ and $D = 0.35$ the output current ripple is identical, so the curves overlap.




(a) Neg. Coupling DC (b) Pos. Coupling DC

(c) Neg. Coupling Fundamental (d) Pos. Coupling Fundamental

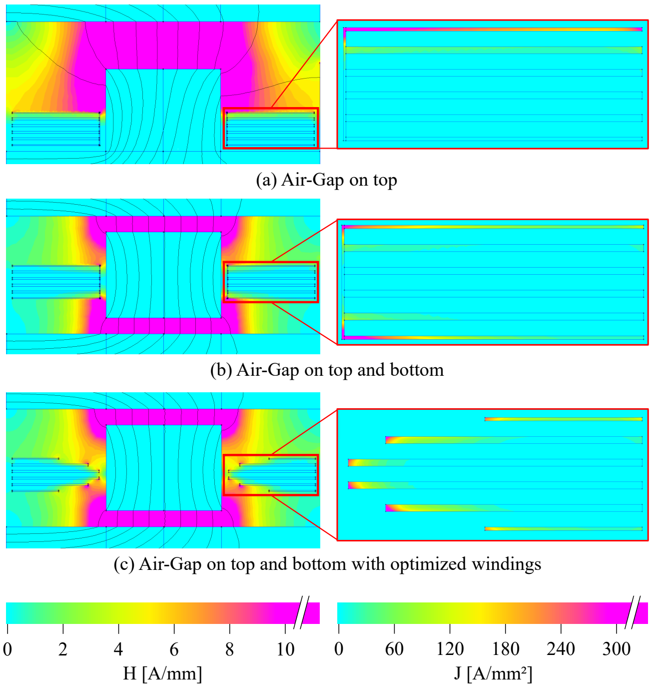
Fig. 6: Flux distribution of the four-pole structure for negative and positive coupling of otherwise identical designs for a DC phase current of 40 A and 100 A of ripple.

a fringing field in the middle of the windings. This partly compensates the internal proximity effect and draws current to the center, reducing losses. The main downside of this design is the significant external field on top and bottom of the inductor, making cooling difficult as no conductive material can be placed on top or bottom of the core.

All of our designs use a single turn with all six layers in parallel, which differs significantly from the original designs. Designs with more than one turn were not considered because the desired inductance and air gap could not be achieved in that case due to fringing and flux leakage between the pillars.

*b) Material limitations:* TDK's PC200 was selected for this design as it exhibits very low loss in the range of 1 MHz to 4 MHz. However, the performance of PC200 significantly degrades for a field strength $H_{\mathrm{dc}} > 50\,\mathrm{A/m}$, and at $H_{\mathrm{dc}} > 100\,\mathrm{A/m}$ its losses double [17]. Therefore, the inductor was designed with a maximum $H_{\mathrm{dc}}$ of $40\,\mathrm{A/m}$ to have some margin.

*c) Positive versus negative coupling:* Equations (5) and (6) are plotted in Fig. 5 for positive and negative air gaps. A strong positive coupling decreases $\Delta I_{\mathrm{out}}$ while increasing $\Delta I_{\mathrm{leg}}$. Overall, positive coupling is slightly beneficial from an electrical point of view as it reduces the output ripple. The higher $\Delta I_{\mathrm{leg}}$ requires a larger $L_{\mathrm{self}}$ resulting in even lower




(a) Air-Gap on top

(b) Air-Gap on top and bottom

(c) Air-Gap on top and bottom with optimized windings

Fig. 7: Field and current distribution for different approaches. Because all layers are in parallel the current crowds in areas of large field strength. By splitting the gap, the field is distributed more evenly and the current distribution improves. Utilizing a smaller width for the outer layers, i.e. moving the copper away from the high-flux areas distributes the current even better.

$\Delta I_{\mathrm{out}}$. However, there is a significant difference in the flux distribution as shown in Fig. 6: For negative coupling, the two windings generate an opposing magnetomotive force at DC, resulting in a low flux that circulates through the outer air gaps. For positive coupling, the two magnetomotive forces are driving a DC-flux in the same direction causing a flux that only circulates between the two windings where the reluctance is much lower. The peak flux-density and consequently also the field strength in the core is approximately twice as high for positive coupling compared to negative coupling. This also means, the cross-sectional areas need to be twice as large for positive coupling as the core has to be designed with respect to the $H_{\mathrm{dc}}$ limit. Therefore, negative coupling was selected for this application[1].

*d) Simulation Process:* For the simulations, the open-source 2D FEM software FEMM was used due to its high speed and straightforward integration with Python. As the core structures are neither planar nor axisymmetric, the designs were first transformed to a planar structure. All cross-sectional areas are kept the same and the depth of the design is determined by the length of the winding.

*e) Split air gap and optimized windings:* The external proximity effect caused by the air gap plays an important role for the current distribution and consequently the losses of this

[1]Note that for lower frequency materials which are less affected by DC flux, the picture would change: Those designs would likely benefit from positive coupling as the flux density at the fundamental is reduced.

(C)2026IEEJ

TABLE I: Comparison of the size and simulated loss of the optimized pillar and windings for the four-pole design.

| | **Volume** $[\mathrm{cm}^3]$ | **Copper Loss** [W] |
|---|---|---|
| *Original design* | 5.8 | 7.0 |
| *Air gap on top and bottom* | 5.5 | 5.7 |
| *Air gap on top and bottom with optimized windings* | 5.3 | 5.0 |

TABLE II: Comparison of the size and simulated loss of the different core structures for 1.5 MHz respecting the $H_{\mathrm{dc}}$ limit. The four-pole structure shows the lowest losses and volume.

| | **Total Area** $[\mathrm{cm}^2]$ | **Height** [cm] | **Volume** $[\mathrm{cm}^3]$ | **Copper Loss** [W] |
|---|---|---|---|---|
| *Four-Pole* | 5.5 | 0.9 | 5.3 | 5.0 |
| *Double PQ* | 6.2 | 0.9 | 5.5 | 5.2 |
| *Vertical gap* | 8.0 | 0.7 | 5.8 | 5.6 |
| *EE core* | 8.1 | 1.1 | 8.9 | 6.0 |

high-frequency inductor. The field strength at the top of the windings is large and almost zero at the bottom as shown in Fig. 7. As a result, the AC current is only flowing in the top layers. By centering the pillar vertically in the window such that there is an equal air gap on the top and on the bottom, the field strength is distributed much more uniformly and current flows in the top and bottom layers. This results in a 20 % reduction in losses, which can be reduced even further by using optimized curved windings as shown in Table I.

*f) Comparison of Core Structures:* Table II compares the different core structures for a design with $f_\mathrm{s} = 1.5\,\mathrm{MHz}$ at $\overline{i_\mathrm{out}} = 80\,\mathrm{A}$ and $k = -0.3$. This coupling factor was chosen because it allows the use of small air gaps at the pillar which reduces copper losses due to the smaller external field while still not increasing $\Delta I_\mathrm{out}$ too much. The cross-sectional areas were designed to result in $H_\mathrm{dc} \leq 40\,\mathrm{A/m}$. All designs except the one with vertical gap use the optimized curved windings. The four-pole structure showed both the lowest losses and the lowest volume and was chosen for the implementation. The final design of the four-pole structure with optimized windings is shown in Fig. 8.

## IV. Experimental prototype

The assembled prototype is shown in Fig. 9. The boxed volume of just $12.4\,\mathrm{cm}^3$ results in a power density of $80\,\mathrm{kW/L}$ $(1300\,\mathrm{W/in}^3)$. Its main hardware components are listed in Table III. Two parallel low-side transistors per phase are used as the converter operates at a low duty cycle. To minimize the

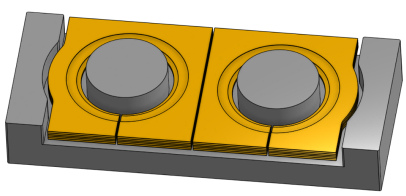


Fig. 8: Final design of the four-pole inductor; top piece not shown.

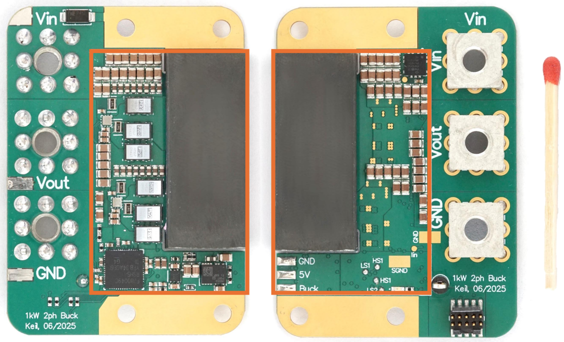
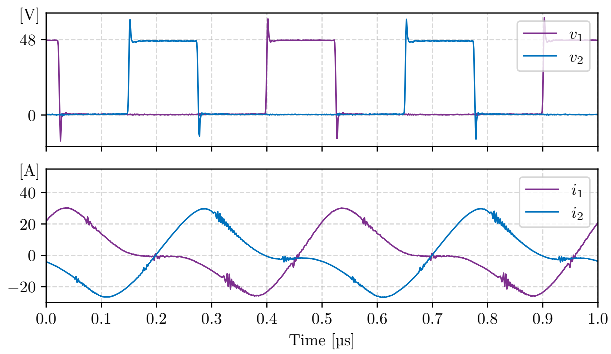


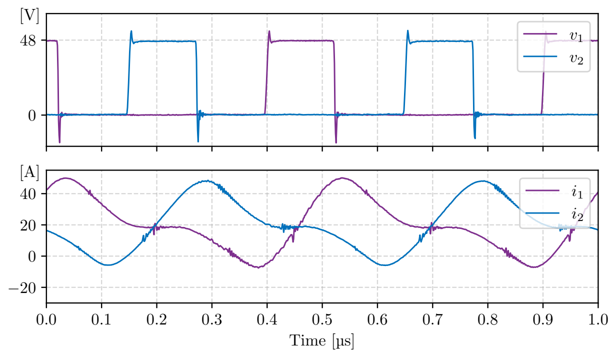

Fig. 9: Fully assembled prototype (without heatspreader). The converter area shown in orange is 45 x 29 x 9.5 mm. Outside of this area only the connectors for power and programming are placed as well as a protection diode and the programming resistors.

TABLE III: Overview of the main hardware components.

| **Component** | **Part** |
|---|---|
| $C_\mathrm{in}$ | 44x 2.2 µF 100 V X7R (20 µF at 48 V) |
| $C_\mathrm{out}$ | 16x 2.2 µF 100 V X7R (33 µF at 12 V) |
| GaN-FETs | Infineon IGC025S08S1 |
| Gate Driver | Analog Devices LT8418 |
| Microcontroller | Texas Instruments F280049C |
| Aux. Power Supply | Texas Instruments TPSM365R6 |
| Current Sensor | Allegro ACS37220 |




(a) $i_\mathrm{out} = 0\,\mathrm{A}$




(b) $i_\mathrm{out} = 40\,\mathrm{A}$

Fig. 10: Measured waveforms at $f_\mathrm{s} = 2\,\mathrm{MHz}$ with a 5 ns dead-time. *Note:* The current-sensor is operating close to its maximum bandwidth and has deskewing applied.

(C)2026IEEJ

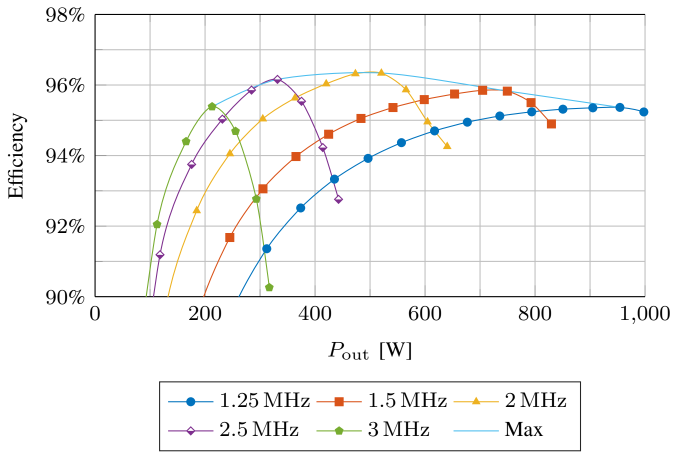



Fig. 11: Efficiency of the prototype for $V_{\mathrm{in}} = 48\,\mathrm{V}$ and $D = 0.25$.

stray inductance, the decoupling capacitors for the half-bridges are connected to the devices with the vertical power loop layout, which uses the first inner layer as a return path [18]. In contrast to other designs, the large onboard capacitance allows the converter to operate without any additional off-board capacitance, while limiting input ripple to $250\,\mathrm{mV}$ and output ripple to $150\,\mathrm{mV}$.

Fig. 10 shows the measured waveforms of the converter. To measure the current, a dedicated PCB was manufactured which includes two Infineon TLE5571 tunnel magnetoresistance (TMR) current sensors (pre-production version). It should be noted that at $2\,\mathrm{MHz}$ these sensors operate close to their bandwidth resulting in slightly distorted waveforms. Nonetheless, the measurement shows that the converter is working as expected: At $i_{\mathrm{out}} = 0\,\mathrm{A}$, the turn-on and turn-off currents are equal, causing equal switching peaks with around $15\,\mathrm{V}$ magnitude. At $i_{\mathrm{out}} = 40\,\mathrm{A}$, the negative current at turn-on is small, causing a slow rise of the switch-node voltage and only a small overshoot of $6\,\mathrm{V}$, while the undershoot at turn-off increases to $18\,\mathrm{V}$. If the current is increased to around $47\,\mathrm{A}$, the converter loses soft switching, causing the significant drop in efficiency beyond $550\,\mathrm{W}$ for $2\,\mathrm{MHz}$ as shown in Fig. 11. The converter's maximum efficiency is $96.3\,\%$ at $550\,\mathrm{W}$. With a power density of $80\,\mathrm{kW/L}$ ($1300\,\mathrm{W/in^3}$), this puts it in the same range as comparable LLC designs [6], [19]–[21]. To use the converter in a real application, a variable switching frequency control is required. The expected resulting efficiency curve is indicated in Fig. 11 as *Max*.

## V. Conclusion

This paper introduced a novel, negatively coupled, single-turn inductor design which enables very high power density and efficiency in a 2-phase buck converter for $48\,\mathrm{V}$ to $12\,\mathrm{V}$ conversion. The challenging demand of a low inductance at high phase current was solved by connecting all layers in parallel with stepped conductor widths to mitigate proximity-effect losses. By using dual air gaps, the current distribution was further improved resulting in a $30\,\%$ reduction in copper losses compared to the standard single air gap design with non-stepped windings.

The effects of positive and negative coupling were evaluated in detail. It was shown that even though negative coupling exhibits higher leg and output current ripple, it allows a $50\,\%$ reduction in core area for cores where the limiting factor is the DC flux. Based on the comparison of four different inductor geometries, the best design was implemented in an experimental prototype which achieves a power density of $80\,\mathrm{kW/L}$ ($1300\,\mathrm{W/in^3}$) for $1\,\mathrm{kW}$ and a peak efficiency of $96.3\,\%$, demonstrating the potential of the proposed inductor design.

## References

[1] A. K. Kumawat and A. K. Thakur, "A Comprehensive Study of Automotive 48-Volt Technology," *International Journal of Mechanical Engineering*, vol. Volume 4, Jan. 2019. doi: 10.14445/23488360/IJME-V4I5P103. [Online]. Available: https://www.internationaljournalssrg.org/IJME/paper-details?Id=73

[2] N. Weitz, S. Ehrlich, P. Freundl, J. Schültzke, and M. März, "A High Frequency Resonant Push-Pull Converter with a Single Integrated Magnetic Component," in *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Mar. 2023. doi: 10.1109/APEC43580.2023.10131366. ISSN 2470-6647 pp. 435–440. [Online]. Available: https://ieeexplore.ieee.org/document/10131366

[3] S. Wang, P. H. Pham, Q. Li, A. Nabih, and P. R. Prakash, "PCB Winding-Based Coupled Inductor for a High-Frequency DC/DC Converter with 99% Efficiency," in *2023 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Mar. 2023. doi: 10.1109/APEC43580.2023.10131439. ISSN 2470-6647 pp. 420–425. [Online]. Available: https://ieeexplore.ieee.org/document/10131439

[4] B. Li, W. Qin, Y. Yang, Q. Li, F. C. Lee, and D. Liu, "A High Frequency High Efficiency GaN Based Bi-Directional 48V/12V Converter with PCB Coupled Inductor for Mild Hybrid Vehicle," in *2018 IEEE 6th Workshop on Wide Bandgap Power Devices and Applications (WiPDA)*, Oct. 2018. doi: 10.1109/WiPDA.2018.8569067 pp. 204–211. [Online]. Available: https://ieeexplore.ieee.org/document/8569067

[5] J. Schäfer, D. Bortis, and J. W. Kolar, "Novel Highly Efficient/Compact Automotive PCB Winding Inductors Based on the Compensating Air-Gap Fringing Field Concept," *IEEE Transactions on Power Electronics*, vol. 35, no. 9, pp. 9617–9631, Sep. 2020. doi: 10.1109/TPEL.2020.2969295 Conference Name: IEEE Transactions on Power Electronics. [Online]. Available: https://ieeexplore.ieee.org/document/8968357

[6] Y. Cai, M. H. Ahmed, Q. Li, and F. C. Lee, "Optimal Design of Megahertz LLC Converter for 48-V Bus Converter Application," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 8, no. 1, pp. 495–505, Mar. 2020. doi: 10.1109/JESTPE.2019.2939469. [Online]. Available: https://ieeexplore.ieee.org/document/8823928

[7] Z. Ouyang and M. A. E. Andersen, "Overview of Planar Magnetic Technology—Fundamental Properties," *IEEE Transactions on Power Electronics*, vol. 29, no. 9, pp. 4888–4900, Sep. 2014. doi: 10.1109/TPEL.2013.2283263. [Online]. Available: https://ieeexplore.ieee.org/document/6607151

[8] W. Liu, Q. Liu, X. Wang, L. Yuan, W. Wu, Q. Qian, and W. Sun, "A High Power Density Coupled Inductor Design for Multi-phase Buck Converters," in *2025 8th International Conference on Energy, Electrical and Power Engineering (CEEPE)*, Apr. 2025. doi: 10.1109/CEEPE64987.2025.11034294 pp. 139–143. [Online]. Available: https://ieeexplore.ieee.org/document/11034294

[9] Z. Saadatizadeh and H. A. Mantooth, "Design Optimization Method of High Power-Density Planar Inductor," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, Oct. 2024. doi: 10.1109/ECCE55643.2024.10861525. ISSN 2329-3748 pp. 3755–3758, iSSN: 2329-3748. [Online]. Available: https://ieeexplore.ieee.org/document/10861525

[10] M. Zheng, Q. Qian, and Q. Liu, "Optimization of Parallel Structure of Planar Inductor Winding," in *2020 21st International Conference on Electronic Packaging Technology (ICEPT)*, Aug. 2020. doi: 10.1109/ICEPT50128.2020.9202487 pp. 1–5. [Online]. Available: https://ieeexplore.ieee.org/document/9202487

(C)2026IEEJ

[11] C. Nan and R. Ayyanar, “A 1 MHz bi-directional soft-switching DC-DC converter with planar coupled inductor for dual voltage automotive systems,” in *2016 IEEE Applied Power Electronics Conference and Exposition (APEC)*, Mar. 2016. doi: 10.1109/APEC.2016.7467908 pp. 432–439. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/7467908

[12] Y. Dong, “Investigation of Multiphase Coupled-Inductor Buck Converters in Point-of-Load Applications,” Jul. 2009. [Online]. Available: http://hdl.handle.net/10919/28469

[13] D. Sha, Y. Zhao, and D. Zhang, “ZVS-Interleaved Synchronous Buck DC–DC Converter With a Coupled Inductor by Varying Switching Frequency and Deadtime,” *IEEE Transactions on Power Electronics*, vol. 37, no. 7, pp. 8190–8198, Jul. 2022. doi: 10.1109/TPEL.2022.3141253 Conference Name: IEEE Transactions on Power Electronics. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/9674799

[14] M. Hua, J. Chen, G. Xu, and H. Wu, “Ultra-thin Coupled Inductor for a GaN-Based CRM Buck Converter,” in *2021 IEEE Workshop on Wide Bandgap Power Devices and Applications in Asia (WiPDA Asia)*, Aug. 2021. doi: 10.1109/WiPDAAsia51810.2021.9656036 pp. 138–142. [Online]. Available: https://ieeexplore.ieee.org/document/9656036/citations#citations

[15] C. Henze, H. Martin, and D. Parsley, “Zero-voltage switching in high frequency power converters using pulse width modulation,” in *APEC ’88 Third Annual IEEE Applied Power Electronics Conference and Exposition*, Feb. 1988. doi: 10.1109/APEC.1988.10548 pp. 33–40. [Online]. Available: https://ieeexplore.ieee.org/document/10548

[16] M. Haider, J. A. Anderson, N. Nain, G. Zulauf, J. W. Kolar, D. Xu, and G. Deboy, “Analytical Calculation of the Residual ZVS Losses of TCM-Operated Single-Phase PFC Rectifiers,” *IEEE Open Journal of Power Electronics*, vol. 2, pp. 250–264, 2021. doi: 10.1109/OJPEL.2021.3058048. [Online]. Available: https://ieeexplore.ieee.org/document/9351687

[17] TDK, “High-Frequency, Low-Loss Ferrite Material PC200.” [Online]. Available: https://product.tdk.com/de/techlibrary/productoverview/ferrite_pc200.html

[18] D. Reusch and J. Strydom, “Understanding the Effect of PCB Layout on Circuit Performance in a High-Frequency Gallium-Nitride-Based Point of Load Converter,” *IEEE Transactions on Power Electronics*, vol. 29, no. 4, pp. 2008–2015, Apr. 2014. doi: 10.1109/TPEL.2013.2266103. [Online]. Available: https://ieeexplore.ieee.org/document/6531683

[19] M. H. Ahmed, A. Nabih, F. C. Lee, and Q. Li, “Low-Loss Integrated Inductor and Transformer Structure and Application in Regulated LLC Converter for 48-V Bus Converter,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 8, no. 1, pp. 589–600, Mar. 2020. doi: 10.1109/JESTPE.2019.2952878. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8895995

[20] M. H. Ahmed, F. C. Lee, Q. Li, and M. d. Rooij, “Design Optimization of Unregulated LLC Converter with Integrated Magnetics for Two-Stage 48V VRM,” in *2019 IEEE Energy Conversion Congress and Exposition (ECCE)*, Sep. 2019. doi: 10.1109/ECCE.2019.8912785. ISSN 2329-3748 pp. 521–528, iSSN: 2329-3748. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8912785

[21] M. H. Ahmed, C. Fei, F. C. Lee, and Q. Li, “48-V Voltage Regulator Module With PCB Winding Matrix Transformer for Future Data Centers,” *IEEE Transactions on Industrial Electronics*, vol. 64, no. 12, pp. 9302–9310, Dec. 2017. doi: 10.1109/TIE.2017.2711519. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/7938363

(C)2026IEEJ